

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 2

Adaptive Classified

OL-T28 · Forming and Solving Equations

With Answers

Questions with full worked solutions

■ 3 pages

FAMILY MAP

Forming Solving Equations

F01

Derive a general expression or formula

1 Q | 3 marks

20 Here is a quadratic equation.

$$ax^2 + 4x + c = 0$$

The solutions of this equation are given by $x = \frac{-4 \pm 2\sqrt{39}}{10}$

Find the value of a and the value of c

Show your working clearly.

$a =$

$c =$

(Total for Question 20 is 3 marks)

Worked solution

November 2025 | Question 20 | 3 marks

Family: Derive a general expression or formula | Variant: General geometric formula derivation

METHOD / WORKING

Worked solution
November 2025 Question 20 3 marks
Family: Derive a general expression or formula Variant: General geometric formula derivation
Method
For
$ax^2 + 4x + c = 0$
the quadratic formula gives
$x = \frac{-4 \pm \sqrt{4^2 - 4ac}}{2a}$
The given solutions are
$x = \frac{-4 \pm 2\sqrt{39}}{10}$
Compare denominators:
$2a = 10$
$a = 5$
Compare the square root parts:
$\sqrt{16 - 4ac} = 2\sqrt{39}$
Square both sides:
$16 - 4ac = 156$
Use $a = 5$:
$16 - 20c = 156$
$-20c = 140$
$c = -7$
Answer: $a = 5, c = -7$
PROVISIONAL WORKED SOLUTION — NOVEMBER 2025 QP-ONLY EXCEPTION

FINAL ANSWER